

Car Price Prediction

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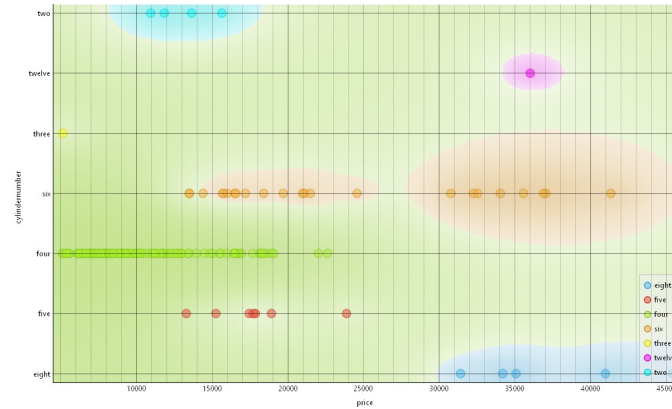
Business Problem

The automobile industry is quite shifting, and a number of elements, such as body style, fuel system, and engine type, affect car prices. Exploratory data analysis (EDA) and machine learning modeling will be used in this research to look into the variables influencing auto prices. The objective is to obtain insights into price fluctuations and create a car pricing prediction model

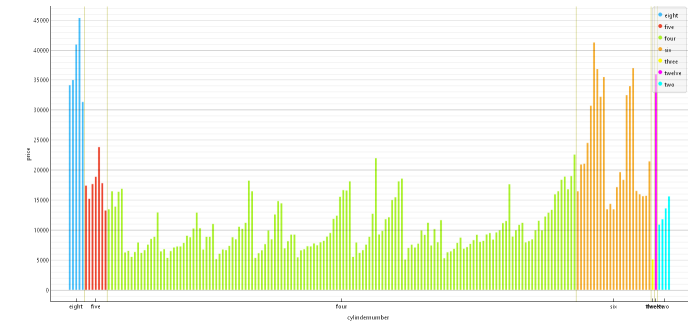
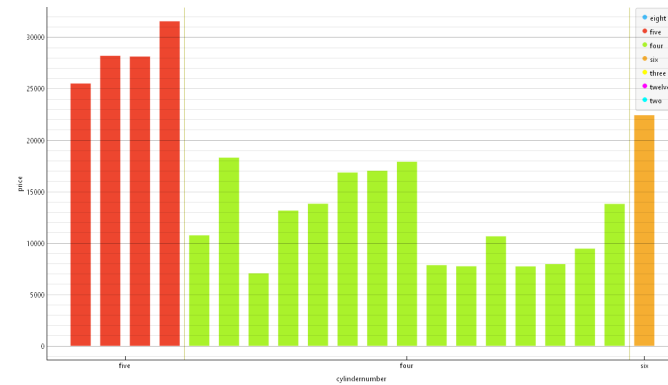
Data Dictionary

	Feature 1	Feature 5	DATA DICTIONARY
1	Car_ID	Unique id of each observation (Interger)	1
2	Symboling	Its assigned insurance risk rating, A value of +3 indicates that the auto is risky, -3 that it is probably pretty safe.(Categorical)	2
3	carCompany	Name of car company (Categorical)	3
4	fueltype	Car fuel type i.e gas or diesel (Categorical)	4
5	aspiration	Aspiration used in a car (Categorical)	5
6	doornumber	Number of doors in a car (Categorical)	6
7	carbody	body of car (Categorical)	7
8	drivewheel	type of drive wheel (Categorical)	8
9	engineLocation	Location of car engine (Categorical)	9
10	wheelbase	Weelbase of car (Numeric)	10
11	carlength	Length of car (Numeric)	11
12	carwidth	Width of car (Numeric)	12
13	carheight	height of car (Numeric)	13
14	curbweight	The weight of a car without occupants or baggage. (Numeric)	14
15	enginetype	Type of engine. (Categorical)	15
16	cylindernumber	cylinder placed in the car (Categorical)	16
17	enginesize	Size of car (Numeric)	17
18	fuelsystem	Fuel system of car (Categorical)	18
19	boreratio	Boreratio of car (Numeric)	19
20	stroke	Stroke or volume inside the engine (Numeric)	20
21	compressionratio	compression ratio of car (Numeric)	21
22	horsepower	Horsepower (Numeric)	22
23	peakrpm	car peak rpm (Numeric)	23
24	citympg	Mileage in city (Numeric)	24
25	highwaympg	Mileage on highway (Numeric)	25
26	price(Dependent variable)	Price of car (Numeric)	26

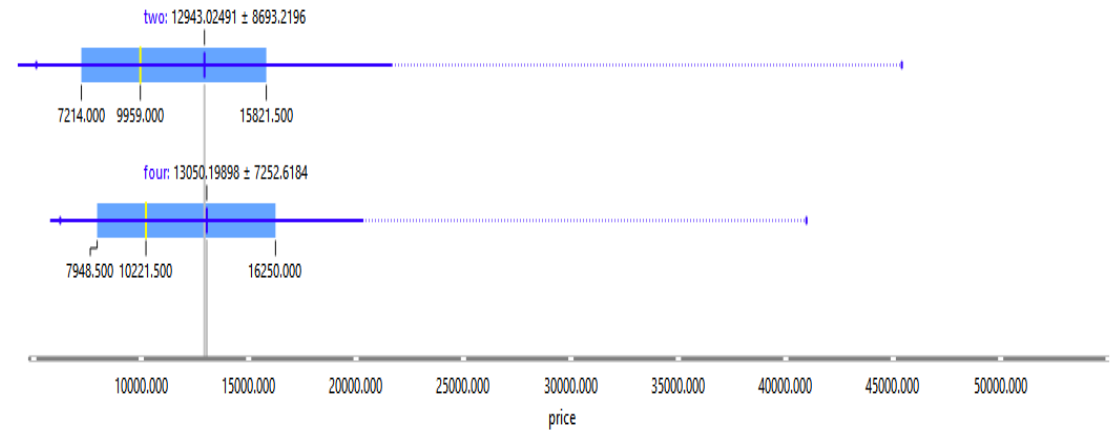
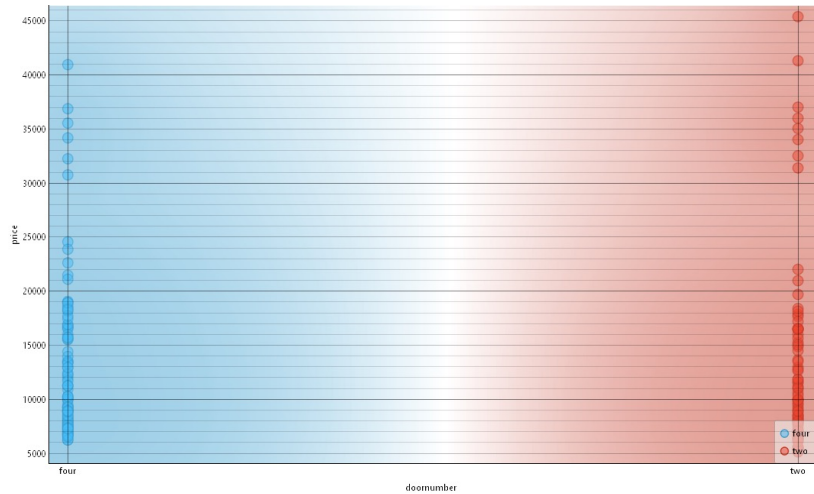
Price Variation by Cylinder Number



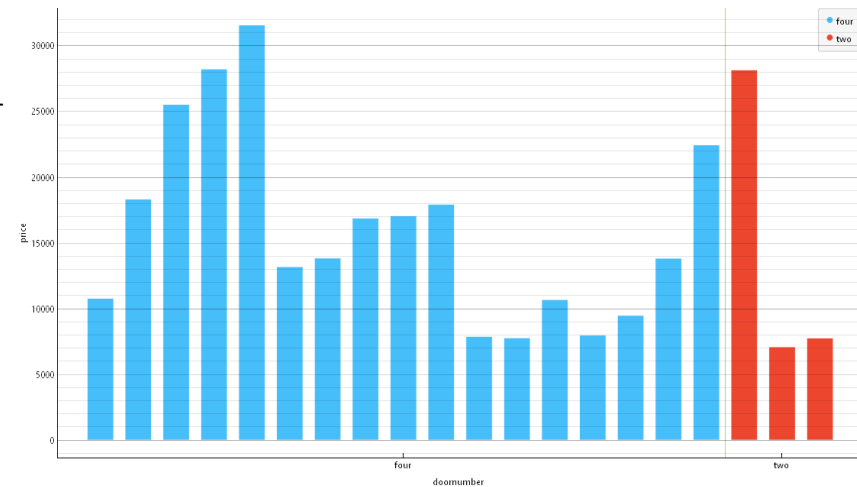
As the number of cylinders increases, the average price of the car also rises.
4-cylinder engines dominate the budget-friendly market.



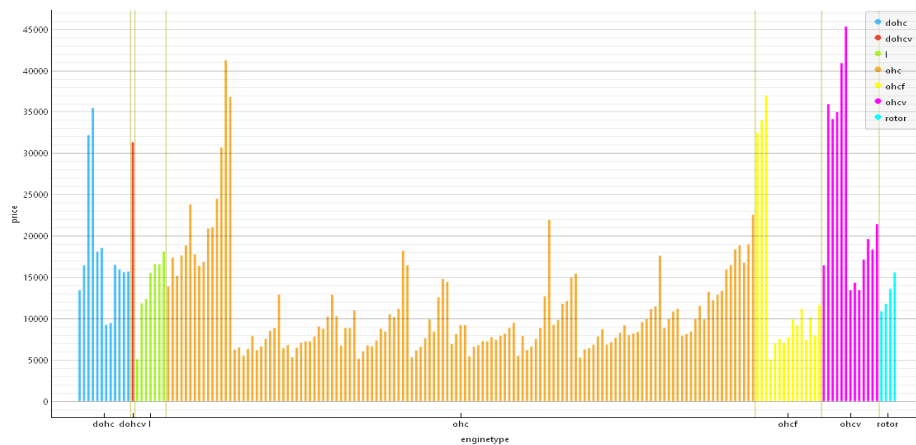
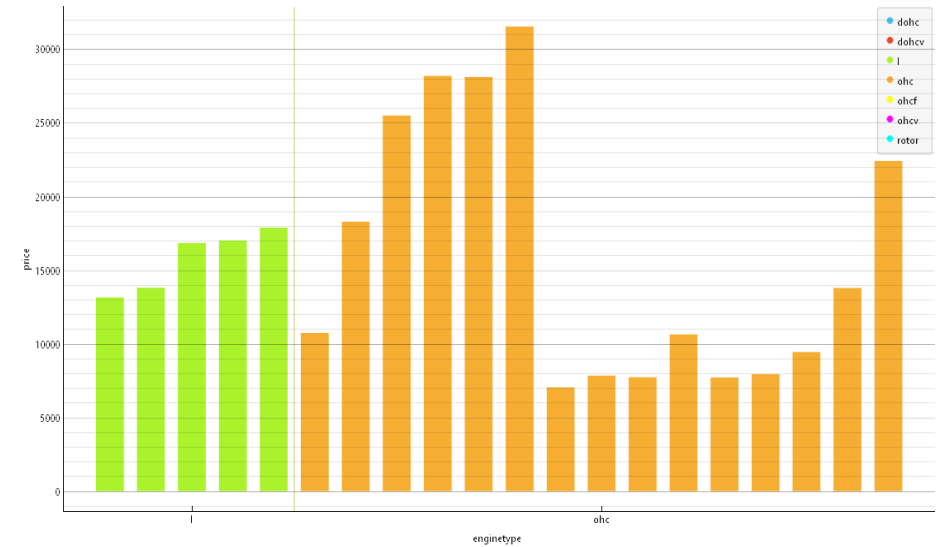
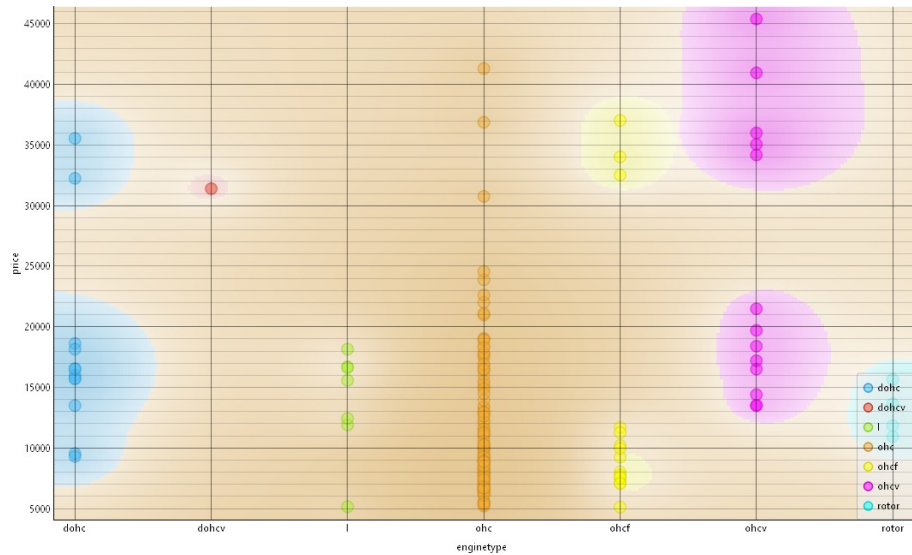
Impact of Number of Doors on Price (diesel Cars)



Gas and diesel cars with 2 and 4 doors have very similar average prices. Diesel 4-door cars show the widest price range, indicating the presence of both budget-friendly and luxury models. Gas 2-door models exhibit more price fluctuations at higher values, suggesting that sports or performance-oriented models drive up variability.



Effect of Engine Type on Price



Significant price variations exist among DOHC, OHCV, and rotor engines.

DOHC (Dual Overhead Camshaft) engines:

Tend to be found in high-performance and luxury vehicles, leading to higher prices.

OHCV (Overhead Camshaft with Valves) engines:

Typically used in powerful, durable cars, often in trucks and performance sedans.

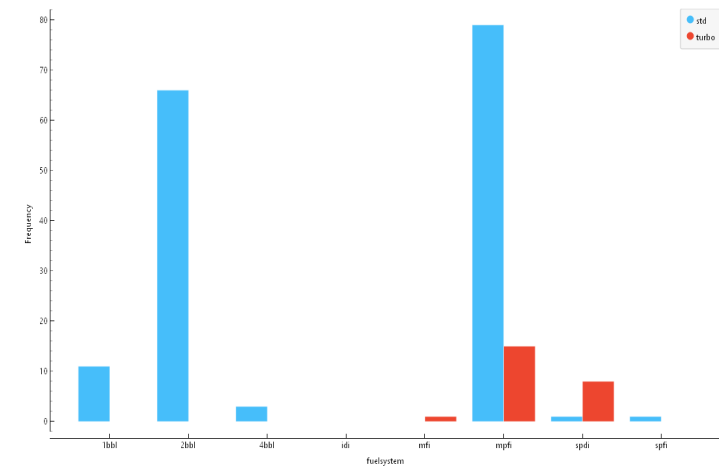
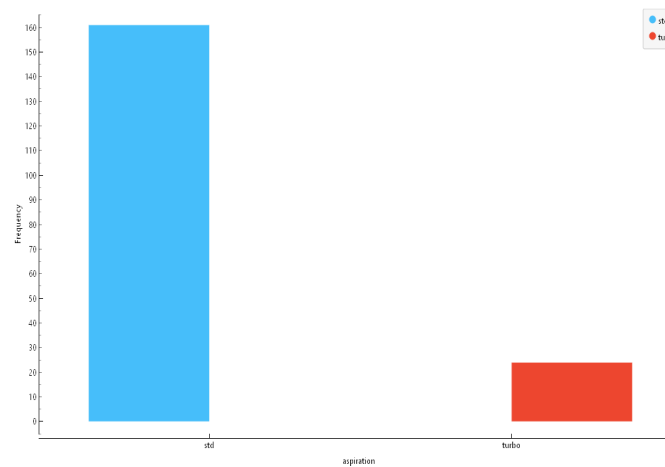
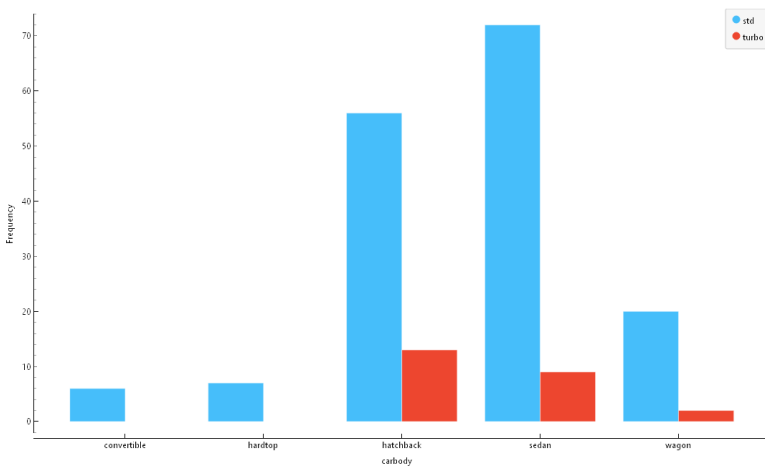
More expensive than simpler engine types.

Rotor (Rotary) Engines:

Less common but often found in specialized sports cars.

Higher prices due to unique design, high-revving capability, and exclusivity.

Influence of Fuel System, Aspiration, and Car Body



Turbocharged gas cars tend to have higher prices compared to standard aspiration.

This is likely due to higher performance, increased horsepower, and better acceleration, making them more desirable.

Turbo engines often appear in sports cars and luxury vehicles, driving up their cost.

MPFI (Multi-Point Fuel Injection) systems dominate the **higher price ranges**.

MPFI systems are more **efficient and precise**, leading to better fuel combustion and performance.

Less advanced fuel injection systems (e.g., carbureted or single-point) are found in **lower-priced** vehicles.

Sedans and Convertibles tend to be more expensive than hatchbacks and wagons.

Feature Statistics

- Car Lengths:
 - Car lengths range from 141.1 to 208.1 inches, with an average of 174.05 inches. This wide range suggests a mix of both compact and full-sized vehicles in the dataset.
- Wheelbase:
 - The average wheelbase is 98.76 inches, with some vehicles extending up to 120.9 inches. This implies the presence of larger vehicles (possibly SUVs or luxury sedans) alongside smaller cars.
- Number of Doors:
 - The median number of doors is four, indicating that most cars are likely sedans. Since coupes and sports cars typically have two doors, their presence is likely but less common.
- Risk Factor Range:
 - The risk factor ranges from -2 to 3, with an average of 0.83.
 - Higher values (closer to 3) suggest riskier vehicles—possibly sports cars or vehicles with high performance.
 - Negative values may indicate safer vehicles, potentially due to better safety features or lower accident rates.

Correlation Analysis



Factors Positively Correlated with Price:

Engine Size (0.87) – The biggest factor influencing price; larger engines generally indicate more powerful and expensive cars.

Horsepower (0.81) – More powerful cars tend to be more expensive.

Curb Weight (0.84) – Heavier cars, often due to luxury features or larger builds, have higher prices.

Wheelbase (0.58, 0.68) – A longer wheelbase is linked to higher prices, possibly due to larger or more stable vehicles.

Car Length (0.76) & Car Width (0.76) – Larger dimensions contribute to higher costs, indicating premium or full-sized vehicles.

Correlations Between Other Features:

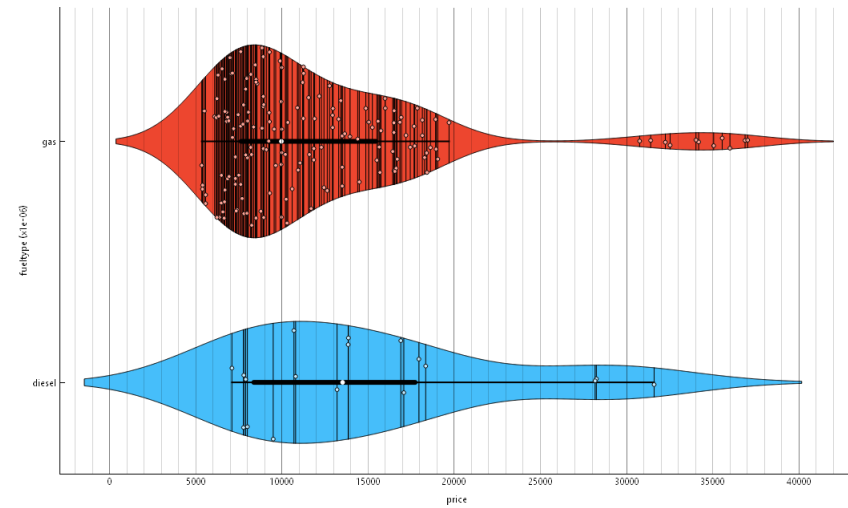
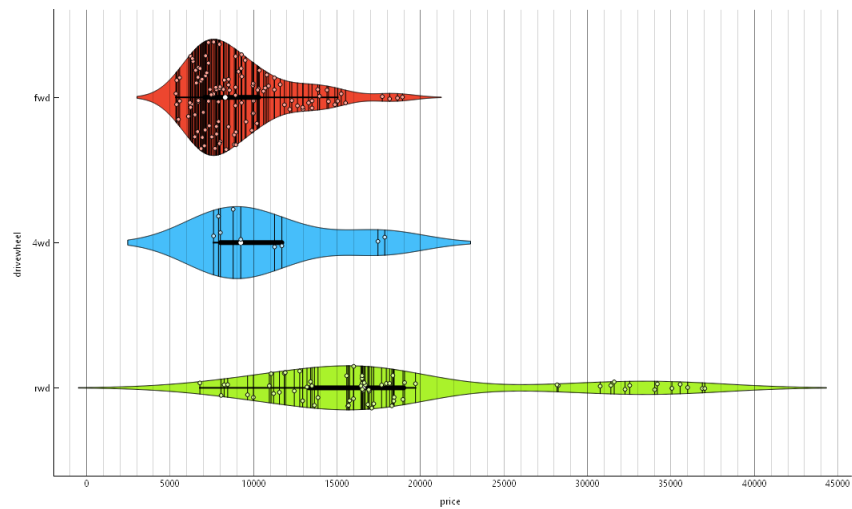
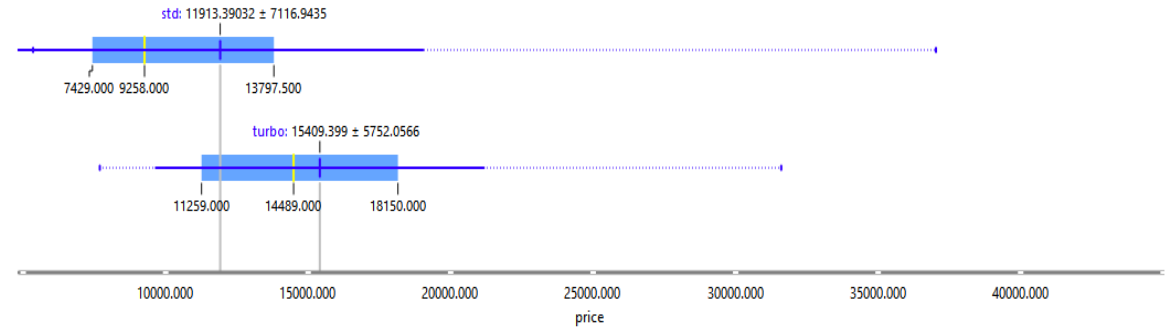
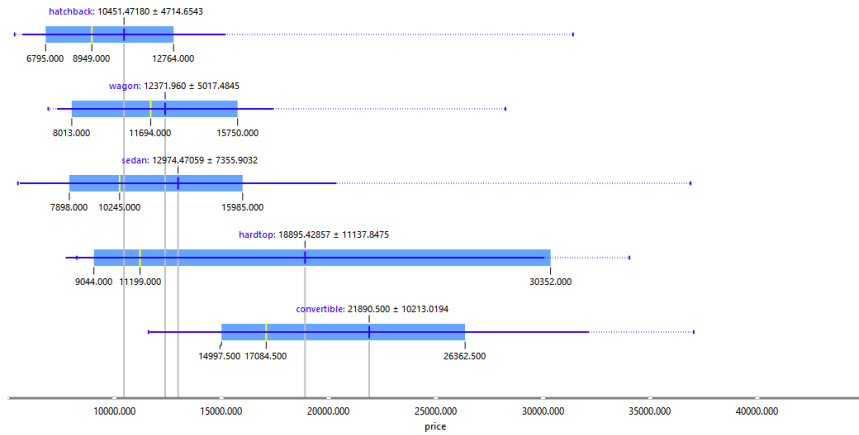
Cylinder Number is strongly correlated with:

Engine Size (0.85) – More cylinders usually mean a bigger engine.

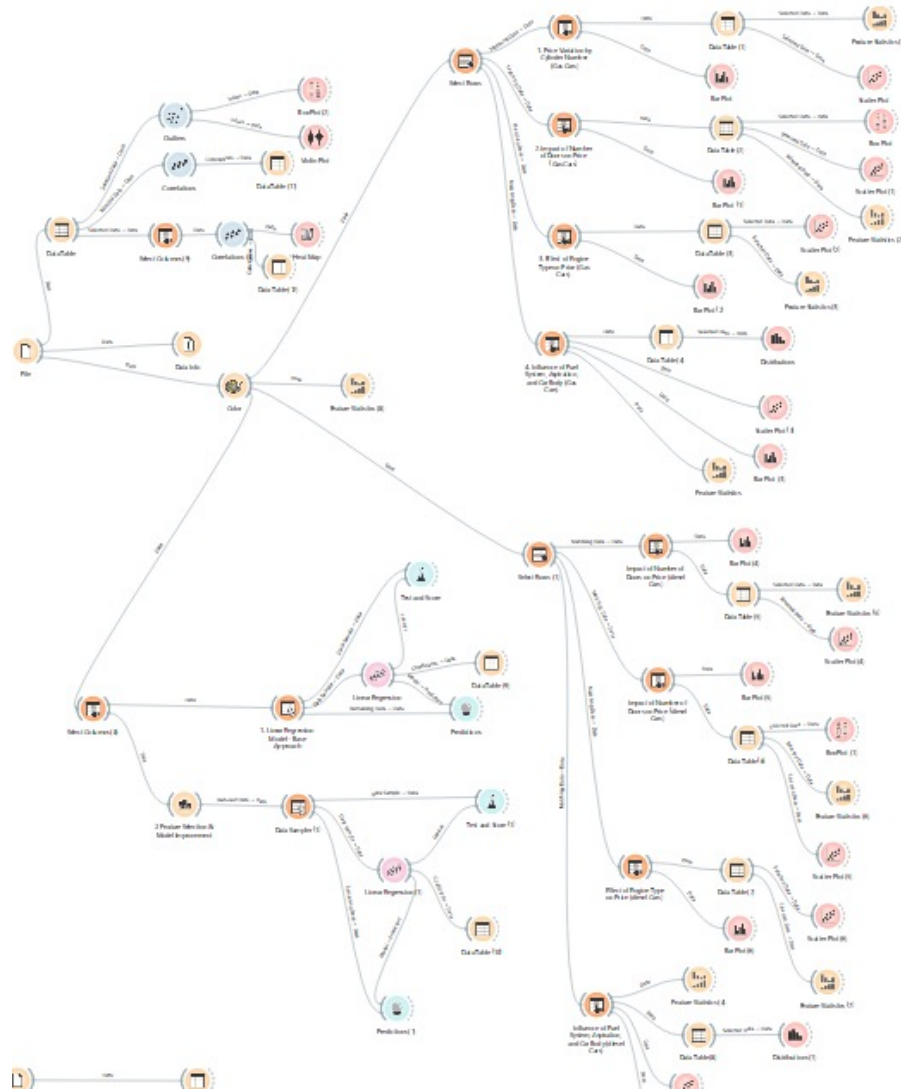
Horsepower (0.75) – More cylinders contribute to greater power.

Curb Weight (0.61) – Larger engines require sturdier, heavier car bodies.

Violin and Box Plot Analysis



Orange Workflow



Final Observations

Engine-Related Features Have the Biggest Impact:

Engine Size, Horsepower, and Curb Weight are the strongest predictors of price.

Aspiration (e.g., turbocharged vs. naturally aspirated engines) also plays a role.

Feature Selection Improved Model Performance:

By removing less relevant features, the model maintained predictive power while achieving a higher R^2 score.

Engine-related attributes (size, horsepower, aspiration) dominate price determination.

Fuel efficiency (MPG) and the number of doors have minimal influence on price.

Luxury and performance cars (higher cylinders, turbocharged engines, premium body types) tend to be in the higher price range.

Brand reputation and luxury features justify higher costs beyond just technical specifications.